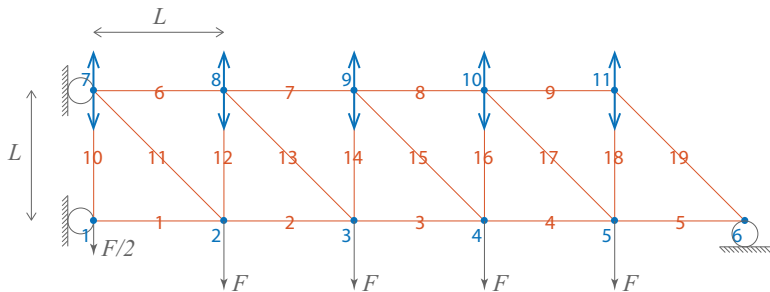


Chapter 6 – Sensitivity analysis



Example: shape optimization of a truss girder



$$L = 1 \text{ m}, \quad E = 210 \text{ GPa}, \quad F = 10 \text{ kN}, \quad A = 0.001 \text{ m}^2$$

- The objective is to minimize the volume of the structure subject to a displacement constraint $u_{\max} = 5 \text{ mm}$ for the downward translation of node 1.
- Design variables: the vertical coordinates of nodes 7 to 11 are allowed to move 0.9 m downwards and 9 m upwards.

Example: shape optimization of a truss girder

- Nested formulation of a structural optimization problem ($\mathbb{SO}_{\text{nf}}$):

$$\begin{aligned} \min_{\mathbf{x}} : \quad & g_0(\mathbf{x}, \mathbf{u}(\mathbf{x})) \\ \text{s.t.} : \quad & g_i(\mathbf{x}, \mathbf{u}(\mathbf{x})) \leq 0, \quad i = 1, \dots, l \\ & x_j^{\min} \leq x_j \leq x_j^{\max}, \quad j = 1, \dots, n \\ \text{where} \quad & \mathbf{u}(\mathbf{x}) = \mathbf{K}^{-1}(\mathbf{x}) \mathbf{F}_p(\mathbf{x}) \end{aligned}$$

or:

$$\begin{aligned} \min_{\mathbf{x}} : \quad & \hat{g}_0(\mathbf{x}) \\ \text{s.t.} : \quad & \hat{g}_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, l \\ & x_j^{\min} \leq x_j \leq x_j^{\max}, \quad j = 1, \dots, n \\ \text{where} \quad & \hat{g}_i(\mathbf{x}) = g_i(\mathbf{x}, \mathbf{u}(\mathbf{x})), \quad i = 0, \dots, l \end{aligned}$$

Note: a distinction is made between g_i and $\hat{g}_i$ to differentiate between the partial derivative $\frac{\partial g_i}{\partial x_j}$ and the total derivative $\frac{dg_i}{dx_j} = \frac{\partial \hat{g}_i}{\partial x_j}$.

Example: shape optimization of a truss girder

- For the truss example:

$$g_0(\mathbf{x}, \mathbf{u}) = V(\mathbf{x})$$

$$g_1(\mathbf{x}, \mathbf{u}) = \frac{\mathbf{l}^T \mathbf{u}}{u_{\max}} - 1 \quad \text{where} \quad \mathbf{l} = [0, \dots, 0, -1, 0, \dots, 0]^T$$

- We need the sensitivities (or total derivatives) $\frac{\partial \hat{g}_i}{\partial x_j}$ of the objective function $\hat{g}_0$ and the constraints $\hat{g}_i$ with respect to all design variables x_j for gradient-based optimization.

6.1 Numerical methods

- Forward difference approximation of $\partial \hat{g}_i / \partial x_j$ at a design $\mathbf{x}^k$:

$$\frac{\partial \hat{g}_i}{\partial x_j} \approx \frac{\hat{g}_i(\mathbf{x}^k + h\mathbf{e}_j) - \hat{g}_i(\mathbf{x}^k)}{h}$$

- Central difference approximation of $\partial \hat{g}_i / \partial x_j$ at a design $\mathbf{x}^k$:

$$\frac{\partial \hat{g}_i}{\partial x_j} \approx \frac{\hat{g}_i(\mathbf{x}^k + h\mathbf{e}_j) - \hat{g}_i(\mathbf{x}^k - h\mathbf{e}_j)}{2h}$$

6.1 Numerical methods

MATLAB code to determine $\frac{\partial \hat{g}_1}{\partial x_j}$ in the truss example (see also ex1a.m):

```
L = nodalvalues(DOF,1.02,-1);
F = nodalvalues(DOF,[1.02 2.02 3.02 4.02 5.02],[-0.5e4 -1e4 -1e4 -1e4 -1e4]);
K = asmkm(Nodes,Elements,Types,Sections,Materials,DOF);
U = K\F;
f = L'*U/umax-1;
h = 1e-5;
dfdx = zeros(1,n);
for k = 1:n
    Nodes1 = Nodes;
    Nodes1(k+6,3) = Nodes1(k+6,3)+h;
    K1 = asmkm(Nodes1,Elements,Types,Sections,Materials,DOF);
    U1 = K1\F;
    f1 = L'*U1/umax-1;
    dfdx(k) = (f1-f)/h;
end
```

6.2 Analytical methods

- In order to obtain analytical expressions for $\partial \hat{g}_i / \partial x_j$ at a design $\mathbf{x}^k$, the chain rule is first applied:

$$\frac{\partial \hat{g}_i}{\partial x_j} = \frac{\partial g_i}{\partial x_j} + \frac{\partial g_i}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial x_j}$$

where $\partial g_j / \partial \mathbf{u}$ is a row matrix and $\partial \mathbf{u} / \partial x_j$ is a column matrix.

- For the truss example:

$$\begin{aligned} \frac{\partial \hat{g}_0}{\partial x_j} &= \frac{\partial g_0}{\partial x_j} + \cancel{\frac{\partial g_0}{\partial \mathbf{u}}} \frac{\partial \mathbf{u}}{\partial x_j} = \frac{\partial V}{\partial x_j} \\ \frac{\partial \hat{g}_1}{\partial x_j} &= \cancel{\frac{\partial g_1}{\partial x_j}} + \frac{\partial g_1}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial x_j} = \frac{\mathbf{l}^T}{u_{\max}} \frac{\partial \mathbf{u}}{\partial x_j} \end{aligned}$$

- How to compute $\frac{\partial \mathbf{u}}{\partial x_j}$?
 - Direct method
 - Adjoint method

6.2.1 Direct method

- The sensitivities $\frac{\partial \mathbf{u}}{\partial x_j}$ are determined as follows:

$$\begin{aligned}\mathbf{K}\mathbf{u} &= \mathbf{F}_p \\ \Rightarrow \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} + \mathbf{K} \frac{\partial \mathbf{u}}{\partial x_j} &= \frac{\partial \mathbf{F}_p}{\partial x_j} \\ \Rightarrow \mathbf{K} \frac{\partial \mathbf{u}}{\partial x_j} &= \frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \\ \Rightarrow \mathbf{K} \frac{\partial \mathbf{u}}{\partial x_j} &= \mathbf{F}_{dj}\end{aligned}$$

where $\mathbf{F}_{dj} = \frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u}$ is called the “pseudo-load” for design variable x_j .

- For the truss example, $\frac{\partial \mathbf{F}_p}{\partial x_j} = 0$, and $\frac{\partial \mathbf{K}}{\partial x_j}$ is computed using StabIl – see further.

6.2.1 Direct method

MATLAB code to determine $\frac{\partial \hat{g}_1}{\partial x_j}$ in the truss example (see also ex1b.m):

```
L = nodalvalues(DOF,1.02,-1);
Fp = nodalvalues(DOF,[1.02 2.02 3.02 4.02 5.02],[-0.5e4 -1e4 -1e4 -1e4 -1e4]);
[K,~,dKdx] = asmkm(Nodes,Elements,Types,Sections,Materials,DOF,dNodesdx,dSectionsdx);
Up = K\Fp;
f = L'*Up/umax-1;
Fd = zeros(nDOF,n);
for k = 1:n
    Fd(:,k) = -dKdx{k}*Up;
end
dUdx = K\Fd;
dfdx = L'*dUdx/umax;
```

6.2.2 Adjoint method

- The expression for $\frac{\partial \mathbf{u}}{\partial x_j}$ is introduced in the expression for $\frac{\partial \hat{g}_i}{\partial x_j}$:

$$\begin{aligned}\frac{\partial \hat{g}_i}{\partial x_j} &= \frac{\partial g_i}{\partial x_j} + \frac{\partial g_i}{\partial \mathbf{u}} \mathbf{K}^{-1} \left(\frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \right) \\ &= \frac{\partial g_i}{\partial x_j} + \boldsymbol{\lambda}_i^T \left(\frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \right)\end{aligned}$$

$$\text{where: } \boldsymbol{\lambda}_i = \mathbf{K}^{-1} \left(\frac{\partial g_i}{\partial \mathbf{u}} \right)^T \Leftrightarrow \mathbf{K} \boldsymbol{\lambda}_i = \left(\frac{\partial g_i}{\partial \mathbf{u}} \right)^T \Leftrightarrow \mathbf{K} \boldsymbol{\lambda}_i = \mathbf{F}_{ai}$$

where $\mathbf{F}_{ai} = \left(\frac{\partial g_i}{\partial \mathbf{u}} \right)^T$ is the so-called “adjoint load” for function g_i
and $\boldsymbol{\lambda}_i$ is the corresponding “adjoint displacement”.

- For the truss example, $\mathbf{F}_{a1} = \left(\frac{\partial g_1}{\partial \mathbf{u}} \right)^T = \frac{1}{u_{\max}}$.

6.2.2 Adjoint method

- Alternative derivation of the adjoint method:

$$\begin{aligned}\hat{g}_i &= g_i + \boldsymbol{\lambda}_i^T (\mathbf{F}_p - \mathbf{K}\mathbf{u}) \\ \frac{\partial \hat{g}_i}{\partial x_j} &= \frac{\partial g_i}{\partial x_j} + \frac{\partial g_i}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial x_j} + \boldsymbol{\lambda}_i^T \frac{\partial \mathbf{F}_p}{\partial x_j} - \boldsymbol{\lambda}_i^T \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} - \boldsymbol{\lambda}_i^T \mathbf{K} \frac{\partial \mathbf{u}}{\partial x_j} \\ &= \frac{\partial g_i}{\partial x_j} + \boldsymbol{\lambda}_i^T \left(\frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \right) + \left(\frac{\partial g_i}{\partial \mathbf{u}} - \boldsymbol{\lambda}_i^T \mathbf{K} \right) \frac{\partial \mathbf{u}}{\partial x_j}\end{aligned}$$

These equations are valid for any adjoint displacement vector $\boldsymbol{\lambda}_i$. We choose the vector $\boldsymbol{\lambda}_i$ such that the last term (with the derivative $\frac{\partial \mathbf{u}}{\partial x_j}$) vanishes, i.e. as the solution of:

$$\mathbf{K}\boldsymbol{\lambda}_i = \left(\frac{\partial g_i}{\partial \mathbf{u}} \right)^T$$

Then:

$$\frac{\partial \hat{g}_i}{\partial x_j} = \frac{\partial g_i}{\partial x_j} + \boldsymbol{\lambda}_i^T \left(\frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \right)$$

6.2.2 Adjoint method

- For the truss example, the adjoint load vector $\mathbf{F}_{a1}$ is given by:

$$\mathbf{F}_{a1} = \left(\frac{\partial g_1}{\partial \mathbf{u}} \right)^T = \frac{\mathbf{l}}{u_{\max}}$$

- For problems involving compliance $C = \mathbf{F}_p^T \mathbf{u}$, the adjoint load vector $\mathbf{F}_a$ needed to determine the sensitivities $\frac{\partial C}{\partial x_j}$ is given by:

$$\mathbf{F}_a = \left(\frac{\partial C}{\partial \mathbf{u}} \right)^T = \mathbf{F}_p$$

As $\mathbf{F}_a = \mathbf{F}_p$, it follows that $\boldsymbol{\lambda} = \mathbf{u}$. The sensitivities $\frac{\partial C}{\partial x_j}$ are then given by:

$$\frac{\partial C}{\partial x_j} = \mathbf{u}^T \left(\frac{\partial \mathbf{F}_p}{\partial x_j} - \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} \right)$$

Or, for problems with design-independent loads:

$$\frac{\partial C}{\partial x_j} = -\mathbf{u}^T \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u}$$

6.2.2 Adjoint method

MATLAB code to determine $\frac{\partial \hat{g}_1}{\partial x_j}$ in the truss example (see also ex1c.m):

```
L = nodalvalues(DOF,1.02,-1);
Fp = nodalvalues(DOF,[1.02 2.02 3.02 4.02 5.02],[-0.5e4 -1e4 -1e4 -1e4 -1e4]);
Fa = L/umax;
F = [Fp Fa];
[K,~,dKdx] = asmkm(Nodes,Elements,Types,Sections,Materials,DOF,dNodesdx,dSectionsdx);
U = K\F;
Up = U(:,1);
Ua = U(:,2);
f = L'*Up/umax-1;
dfdx = zeros(1,n);
for k = 1:n
    dfdx(k) = -Ua'*dKdx{k}*Up;
end
```

6.3 Analytical calculation of pseudo-loads

Focus on $\frac{\partial K}{\partial x_j}$, not on $\frac{\partial F}{\partial x_j}$ (usually trivial)

Stiffness matrix assembly:

$$K = \sum_e C_e^T k_e C_e$$

$$\Rightarrow \frac{\partial K}{\partial x_j} = \sum_e C_e^T \left(\frac{\partial k_e}{\partial x_j} \right) C_e$$

so we need this one.

6.3.1 Bars

Finite elements, part 1:

$$k_e = T_e^T \bar{k}_e T_e$$

where:

$$\bar{k}_e = \frac{EA}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

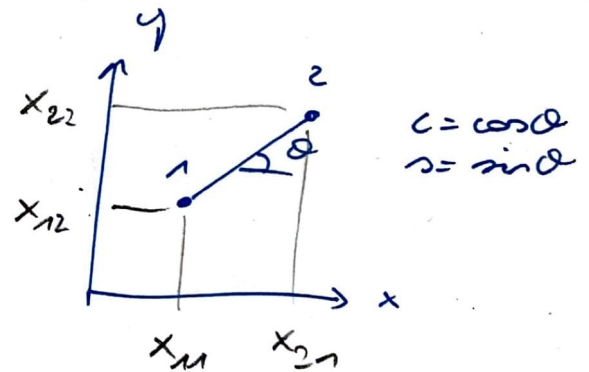
$$T_e = \begin{bmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{bmatrix}$$

where:

$$l = \sqrt{(x_{21} - x_{11})^2 + (x_{22} - x_{12})^2}$$

$$c = \frac{x_{21} - x_{11}}{l} = \frac{x_{21} - x_{11}}{\sqrt{(x_{21} - x_{11})^2 + (x_{22} - x_{12})^2}}$$

$$s = \frac{x_{22} - x_{12}}{l} = \frac{x_{22} - x_{12}}{\sqrt{(x_{21} - x_{11})^2 + (x_{22} - x_{12})^2}}$$



Size derivatives

$$\begin{aligned}\frac{\partial h_e}{\partial A} &= T_e^T \frac{\partial \bar{h}_e}{\partial A} T_e \\ &= T_e^T \left(\frac{E}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right) T_e\end{aligned}$$

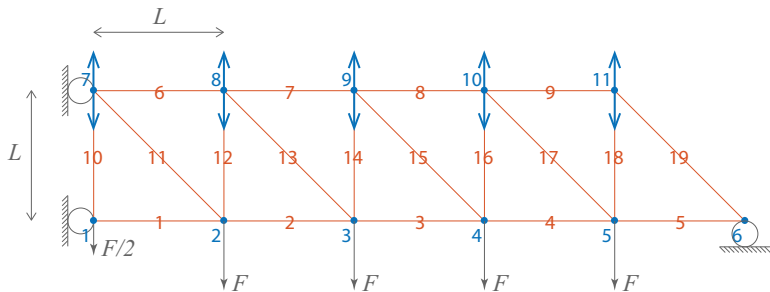
Shape derivatives

$$\frac{\partial h_e}{\partial x_{pq}} = \frac{\partial T_e^T}{\partial x_{pq}} \bar{h}_e T_e + T_e^T \frac{\partial \bar{h}_e}{\partial x_{pq}} T_e + T_e^T \bar{h}_e \frac{\partial T_e}{\partial x_{pq}}$$

→ so we need the derivatives of all terms in T_e and $\bar{h}_e$ w.r.t x_{11}, x_{12}, x_{21} , and x_{22} .
The derivative $\frac{\partial c}{\partial x_{11}}$ is determined as an example:

$$\begin{aligned}\frac{\partial c}{\partial x_{11}} &= \frac{-1}{l} + \frac{1}{2} \frac{x_{21} - x_{11}}{l^3} 2(x_{21} - x_{11}) \\ &= -\frac{1}{l} + \frac{(x_{21} - x_{11})^2}{l^3} \\ &= -\frac{1 - c^2}{l} = -\frac{s^2}{l}\end{aligned}$$

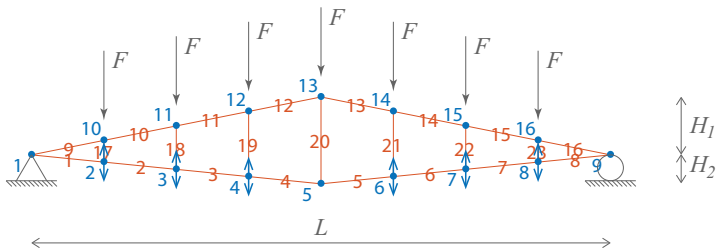
Example: size and shape optimization of a truss girder



$$L = 1 \text{ m}, \quad E = 210 \text{ GPa}, \quad F = 10 \text{ kN}, \quad A = 0.001 \text{ m}^2$$

- Solve the shape optimization problem introduced on slide 94 (ex2a.m).
- Add the bar sections A_j as design variables; allow them to vary between 10^{-7} m^2 and 10^{-3} m^2 (ex2b.m).
- Add displacement constraints $u_{\max} = 5 \text{ mm}$ for the downward translations of nodes 2 to 5 (ex2c.m).
- Reformulate the problem as a minimum compliance problem with a maximum volume of 0.008 m^3 (ex2d.m).

Example: size and shape optimization of a funicular truss



$$L = 20 \text{ m}, H_1 = 2 \text{ m}, H_2 = 1 \text{ m}, E = 30 \text{ GPa}, F = 150 \text{ kN}$$

- Minimize the compliance of the structure subject to a volume constraint $V_{\max} = 2 \text{ m}^3$. The beams have a square cross-section with a side length h that can vary between 0.01 m and 1 m. The vertical coordinates of the nodes 2, 3, 4, 6, 7, and 8 can move between the level of the supports and 2 m lower. The shape of the structure must remain symmetric (ex3.m).

Example: size and shape optimization of a funicular truss



Magazzini Generali, Chiasso, Zwitterland, 1924, Robert Maillart